

Thanks for the clarification — you've installed **Proxmox VE** on an SSD in your **Dell PowerEdge R640**, but now the **system won't boot from that SSD**. The OS was installed successfully, but on reboot, you get:

- > ❌ **"No boot device available"**
- > ❌ **System doesn't recognize the SSD as bootable**
- > ❌ **Proxmox is installed but not booting**

Let's **diagnose and fix** this step by step. This is a **common issue** on Dell servers due to **UEFI, RAID, and bootloader (GRUB) misconfigurations**.

🔍 Root Causes (Most Common on R640)

1. ✅ **GRUB not installed to the correct drive**
2. ✅ **Missing or unmounted EFI System Partition (ESP)**
3. ✅ **Bootting in Legacy mode instead of UEFI**
4. ✅ **Incorrect boot order (PXE/network boot first)**
5. ✅ **RAID virtual disk not marked as bootable**
6. ✅ **Dell firmware bug or boot entry not created**

We'll fix each possibility.

✅ Step 1: Confirm You're in UEFI Mode

Proxmox **requires UEFI** to boot properly. If the system is in **Legacy (BIOS) mode**, it won't see the EFI bootloader.

🛠️ Fix:

1. Reboot → Press **F2** to enter **System Setup**

2. Go to:

...

Boot Settings → Boot Mode → UEFI

...

3. **Disable** "Legacy Option ROMs" if enabled

4. Save and Exit (**F10**)

> ⚠️ If you installed Proxmox in UEFI mode but boot in Legacy — it **won't work**.

✅ Step 2: Check Boot Order

Dell often sets **PXE (network boot)** as the first boot device.

🛠️ Fix:

1. Press **F2** → **Boot Settings** → **Boot Sequence**

2. Look for:

- ✅ `UEFI: SSD-Proxmox` (or similar)

-  `UEFI: Ubuntu` ← This is likely your Proxmox install!

3. Move it to **top of the list**

4. **Disable or move down**:

- `PXE Boot`
- `iPXE`
- `Legacy: ...`

 Save and Exit

 Step 3: Use F11 Boot Menu to Test

Instead of relying on boot order, use **One-Time Boot**:

1. Reboot → Press **F11** during POST

2. In **Boot Manager**, look for:

- `UEFI: SSD-Proxmox`
- `UEFI: Ubuntu`
- `UEFI: Built-in EFI Shell`

➤ Select **"UEFI: Ubuntu"** or your SSD

 If it boots now — great! Your OS is fine. Just fix the **default boot order** (Step 2).

✅ Step 4: Verify RAID Virtual Disk is Healthy

If the RAID array isn't healthy, the system won't boot.

🔧 Check in PERC:

1. Reboot → Press **Ctrl+R** during boot
2. In **PERC BIOS**:
 - Check **Controller Status**: `Optimal`
 - Expand **Virtual Disks**:
 - Your Proxmox disk (e.g., `SSD-Proxmox`) → Status: `Optimal`
 - Expand **Physical Disks**:
 - SSD 1 → Status: `Online`

❌ If degraded or missing → recreate the RAID 0 array.

✅ Step 5: Reinstall GRUB Bootloader (Critical)

Most likely: **GRUB wasn't installed to the EFI partition**.

Let's fix it using the **Proxmox installer USB**.

🛠 Step 5.1: Boot from Proxmox USB

1. Insert USB
2. Press ****F11**** → Boot from ****UEFI: Proxmox USB****

Wait for installer to load.

🛠 Step 5.2: Open Console

On the installer screen, press:

`, `

Ctrl + Alt + F2

`, `

You'll get a shell.

🛠 Step 5.3: Identify Your Drives

Run:

`` `bash

```
lsblk -f
```

```
`,`,`
```

You should see something like:

```
`,`,`
```

```
NAME FSTYPE LABEL MOUNTPOINT
```

```
sda
```

```
└─sda1 vfat EFI /boot/efi ← EFI partition
```

```
└─sda2 ext4 root / ← Proxmox root
```

```
└─sda3 swap swap
```

```
`,`,`
```

> ⚠ If `/boot/efi` is missing or not mounted → GRUB can't boot.

```
---
```

🛠 Step 5.4: Mount Installed System

Replace `/dev/sda2` and `/dev/sda1` with your actual partitions.

```
`,`,`bash
```

```
# Mount root
```

```
mount /dev/sda2 /mnt
```

```
# Mount boot/efi
```

```
mkdir -p /mnt/boot/efi
```

```
mount /dev/sda1 /mnt/boot/efi
```

```
# Bind other required filesystems
```

```
mount --bind /dev /mnt/dev
```

```
mount --bind /proc /mnt/proc
```

```
mount --bind /sys /mnt/sys
```

```
mount --bind /run /mnt/run
```

```
```
```

```

```

```
🛠 Step 5.5: Chroot into Installed System
```

```
```bash
```

```
chroot /mnt
```

```
```
```

Now you're inside your installed Proxmox system.

```

```

```
🛠 Step 5.6: Reinstall GRUB
```

```
```bash
```

```
# Install GRUB to EFI
```

```
grub-install --target=x86_64-efi --efi-directory=/boot/efi --bootloader-id=proxmox --recheck
```

```
# Update GRUB config
```

```
update-grub
```

```
# Rebuild initramfs
```

```
update-initramfs -u -k all
```

```
```
```

✅ Success if no errors.

```

```

### 🛠 Step 5.7: Exit and Reboot

```
```bash
```

```
exit
```

```
umount -R /mnt
```

```
reboot
```

```
```
```

> Remove USB when rebooting

```

```

## ✅ Step 6: Verify EFI Partition Exists



If there's **no EFI partition**, GRUB can't install.

### During Reinstall:

- Choose **"Ext4"** (not ZFS or XFS)
- Let Proxmox create **default partitions**:
  - 512MB EFI (FAT32)
  - Rest for root

Avoid "custom" partitioning unless you know what you're doing.

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##  Step 7: Alternative — Use AHCI Mode (Skip RAID)

If PERC RAID keeps causing boot issues:

###  Switch to AHCI:

1. Press **F2** → **System Setup**

2. Go to:

...

Storage → SATA Operation → AHCI

...

3. Save and Exit

Now:

- Drives appear as `/dev/sda`, `/dev/sdb`, etc.

- Install Proxmox directly to SSD 1
- GRUB installs cleanly

> ⚠️ You lose RAID features, but gain reliability.

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## ## ✅ Step 8: Check iDRAC & Firmware

### ### Update Firmware:

1. Press **F10** at boot → **Dell Lifecycle Controller**
2. Update:
  - BIOS
  - PERC Controller
  - Drive Firmware

Old firmware can cause boot issues.

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## ## ✅ Final Checklist

| ✅ Done? | Task |

|-----|-----|

| ☐ | Boot Mode = **UEFI** |

| ☐ | Boot Order = **SSD first (UEFI)** |

| ☐ | Disabled **\*\*PXE/Network Boot\*\*** |  
| ☐ | RAID Virtual Disk = **\*\*Optimal\*\*** |  
| ☐ | EFI partition exists and mounted |  
| ☐ | GRUB reinstalled via chroot |  
| ☐ | `update-grub` and `grub-install` ran without error |

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## 💬 Still Not Working?

Please run this in the installer console (**\*\*Ctrl+Alt+F2\*\***) and paste the output:

```
```bash
```

```
lsblk -f
```

```
efibootmgr -v
```

```
```
```

I'll tell you:

- If EFI is missing
- If GRUB entry exists
- Which drive to target

We **\*\*will get it working\*\*** — don't worry! 🙌